Roll No.

Total No. of Questions : 07

Total No. of Pages :02

BCA (Sem.–5)

PROGRAMMING IN JAVA

Subject Code :UGCA-1932

M.Code :90315

Date of Examination:29-11-2025

Time : 3 Hrs.

Max. Marks : 60

INSTRUCTIONS TO CANDIDATES :

1. SECTION-A is COMPULSORY consisting of TEN questions carrying TWO marks each.
2. SECTION-B contains SIX questions carrying TEN marks each and students have to attempt any FOUR questions.

SECTION-A

1. **Write briefly:**

a) List four features of Java that support-platform independence and security.

b) What is the JRE and how does it differ from the JDK?

c) Write the general form of an if—else if ladder and state one use-case.

d) When would you prefer switch over multiple if-else statements? Mention two constraints.

e) What do break and continue do in nested loops?

f) Define constructor and state two properties of constructors in Java.

g) What is the difference between String and StringBuffer?

h) Explain primitive data types.

i) Name any four stream classes (byte/character) and one usage of each.

j) Write the basic life-cycle methods of an Applet.

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SECTION-B

2 Explain looping constructs, including loop control (break, continue, labeled break). Provide short examples that illustrate each.

3. Describe classes and objects in Java: access modifiers, passing arguments, constructors (default/parameterized/overloaded), this, static members, and garbage collection.

4. Explain inheritance and polymorphism: single inheritance, overriding, super and constructor chaining, abstract classes *vs* interfaces, access specifiers, and packages (creation, access, and import/static import). Include small examples.

5. Explain multithreading in Java: thread creation (Thread *vs* Runnable), scheduler, priorities, joining and synchronization (intrinsic locks synchronized methods/blocks).

6. Write notes on Java I/O: stream hierarchy (byte *vs* character, buffered/filtered), File/Path, random access file; safe I/O with try-with-resources. Briefly add Applet graphics and event handling overview.

7 Design classes Account and SavingsAccount with methods deposit, withdraw, and interest calculation. Demonstrate overriding and use of super.

***NOTE* : Disclosure of Identity by writing Mobile No. or Marking of passing request on any paper of Answer Sheet will lead to UMC against the Student.**

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